

45.1 Why model more than the mean

A fitted regression is usually reported as a curve with a band around it. The curve estimates $E(Y \mid \mathbf{x})$; the band says how far individual responses fall from it. In every model this book has built, the curve has been the object of study and the band an afterthought, computed from one variance parameter, one variance function or one sandwich correction. That division of labour is a modelling decision, and often a bad one.

45.1.1 What the mean model promises

Take the model of [Chapter 5](#) in its strongest form: $Y = \mathbf{x}^\top \boldsymbol{\beta} + \varepsilon$ with ε independent of $\mathbf{x}$. Write

$$Q_Y(\tau \mid \mathbf{x}) = \inf\{q : P(Y \leq q \mid \mathbf{x}) \geq \tau\}$$

for the conditional quantile function. A homoscedastic model makes it a rigid object: all its level curves are one curve, shifted.

Lemma 45.1.1. Quantiles follow monotone maps

Let Y be a random variable and h a strictly increasing, left-continuous function on the range of Y . Then $Q_{h(Y)}(\tau) = h(Q_Y(\tau))$ for every $\tau \in (0, 1)$.

Proof. Write F for the distribution function of Y and $q_0 = Q_Y(\tau)$. Strict monotonicity gives $\{y : h(y) \leq h(q_0)\} = \{y : y \leq q_0\}$, so $P\{h(Y) \leq h(q_0)\} = F(q_0) \geq \tau$ and $h(q_0)$ is admissible in the infimum defining $Q_{h(Y)}(\tau)$. For the reverse inequality the infimum runs over all real c , not only over values of h , and that is where left-continuity is needed. Take $c < h(q_0)$ and put $b = \sup\{y : h(y) \leq c\}$; if that set is empty then $P\{h(Y) \leq c\} = 0 < \tau$ at once. Otherwise $h(b) = \lim_{y \uparrow b} h(y) \leq c$ by left-continuity, so the set is $(-\infty, b]$ and $P\{h(Y) \leq c\} = F(b)$; and $h(b) \leq c < h(q_0)$ forces $b < q_0$, whence $F(b) < \tau$ by the definition of q_0 . No $c < h(q_0)$ is admissible, so the two infima agree. ■

Proposition 45.1.2. What a mean model does and does not say

Let $\mathbf{x}$ be a covariate vector and ε a random variable with quantile function q_ε , independent of $\mathbf{x}$.

- (a) **Location model.** If $Y = \mathbf{x}^\top \boldsymbol{\beta} + \varepsilon$, then $Q_Y(\tau \mid \mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta} + q_\varepsilon(\tau)$. Every conditional quantile curve is the regression curve shifted by a constant, so the effect of a covariate on the τ -quantile is $\boldsymbol{\beta}$, the same for every τ , and the conditional distribution has the same shape and the same spread everywhere.
- (b) **Location-scale model.** If $Y = \mathbf{x}^\top \boldsymbol{\beta} + \exp(\mathbf{x}^\top \boldsymbol{\alpha})\varepsilon$, then

$$Q_Y(\tau \mid \mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta} + \exp(\mathbf{x}^\top \boldsymbol{\alpha}) q_\varepsilon(\tau), \quad (45.1.1)$$

so $\partial Q_Y(\tau \mid \mathbf{x}) / \partial \mathbf{x} = \boldsymbol{\beta} + q_\varepsilon(\tau) \exp(\mathbf{x}^\top \boldsymbol{\alpha}) \boldsymbol{\alpha}$ does depend on τ . The quantile curves are no longer parallel, but they are still generated by one curve q_ε stretched by one positive function.

- (c) **Beyond scale.** In (b) the standardized response $\{Y - \mathbf{x}^\top \boldsymbol{\beta}\} \exp(-\mathbf{x}^\top \boldsymbol{\alpha})$ has a distri-

bution free of $\mathbf{x}$. Any feature of the conditional law that is invariant under location and scale changes — skewness, kurtosis, the ratio of any two central moments of the same degree — is therefore constant in $\mathbf{x}$. A model of the mean and the variance cannot represent a conditional distribution whose *shape* moves with $\mathbf{x}$.

Proof. Parts (a) and (b) are [Lemma 45.1.1](#) applied conditionally on $\mathbf{x}$ to the strictly increasing map $h(e) = \mathbf{x}^\top \boldsymbol{\beta} + ce$ with $c = 1$ and $c = \exp(\mathbf{x}^\top \boldsymbol{\alpha}) > 0$; the derivative in (b) is immediate from [Eq. 45.1.1](#). For (c), write $Z = \{Y - \mathbf{x}^\top \boldsymbol{\beta}\} \exp(-\mathbf{x}^\top \boldsymbol{\alpha})$. By construction $Z = \varepsilon$, whose law does not involve $\mathbf{x}$, and any location-scale invariant functional of the conditional law of Y equals the same functional of the law of Z . ■

In plain words: quoting one residual standard deviation asserts that the poorest household's food budget is as predictable as the richest one's, and that a covariate moving the median moves the tenth and ninetieth percentiles by exactly the same amount. Those are claims about the world, rarely examined and often false.

45.1.2 Heteroscedasticity is a finding

[Chapter 21](#) treated non-constant variance as a problem to be managed: test for it ([Theorem 21.2.2](#)), fix the standard errors ([Definition 21.4.1](#)), or reweight ([Theorem 21.3.1](#)). All three treat $\text{Var}(Y \mid \mathbf{x})$ as a nuisance, yet the variance function is often the more interesting half of the model. If the spread of insurance claims grows faster than their mean, that is the fact the insurer is pricing; if the variability of a child's height at a given age defines the reference band, the band *is* the model.

Example. Engel's budgets, one more time

Engel's 1857 survey of 235 Belgian working-class households records annual income and annual food expenditure; the same data supported the diagnostics of [Chapter 20](#) and the tests of [Section 21.2](#). Least squares gives a slope of 0.4852 and a single residual standard deviation of 114.1 francs. Split the households into thirds by income: the residual standard deviations within the thirds are 48.6, 74.8 and 171.6 francs, a ratio of 3.53 from the poorest third to the richest.

The consequence for a prediction interval is immediate. The nominal 95% interval from the homoscedastic normal model contains 1.000 of the poorest third, 1.000 of the middle third and 0.911 of the richest — far too wide where the data are tight and too narrow where they are loose — while the overall figure of 0.970 looks respectable. Averaging over the design hides the error that matters to each household.

Shape moves too. The quartile coefficient of skewness, $(q_{0.75} + q_{0.25} - 2q_{0.5}) / (q_{0.75} - q_{0.25})$, is -0.23 for the residuals of the poorest fifth of households and 0.40 for the richest fifth, and the interquartile range grows by a factor of 3.75 between them ([Figure 45.1.1](#)).

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
data = sm.datasets.engel.load_pandas().data
```

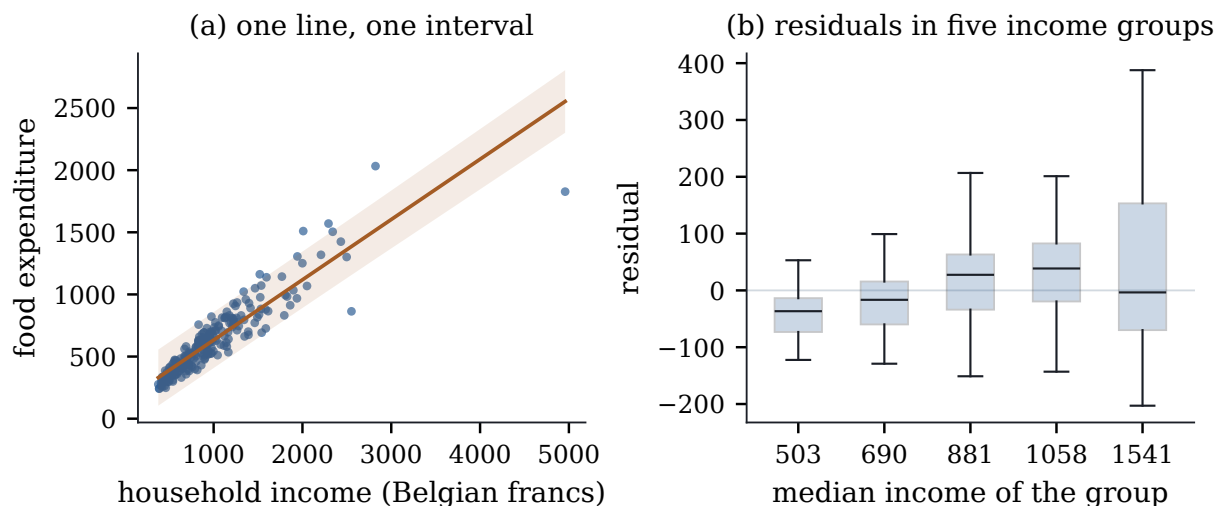


Figure 45.1.1: Engel's budget survey. (a) The least squares line with the 95% prediction band of a homoscedastic normal model. (b) Residuals by income fifth.

```
income = data["income"].to_numpy()
food = data["foodexp"].to_numpy()
n = len(income)
X = np.column_stack([np.ones(n), income])

beta, *_ = np.linalg.lstsq(X, food, rcond=None)
resid = food - X @ beta
sigma = np.sqrt(resid @ resid / (n - 2))          # one number for every household

# a 95% prediction interval from the homoscedastic normal model
t = stats.t.ppf(0.975, n - 2)
lev = np.einsum("ij,jk,ik->i", X, np.linalg.inv(X.T @ X), X)
half = t * sigma * np.sqrt(1 + lev)
inside = (food >= X @ beta - half) & (food <= X @ beta + half)

# split the households into income thirds and look at the spread in each
edges = np.quantile(income, [1 / 3, 2 / 3])
third = np.digitize(income, edges)
spread = [resid[third == k].std(ddof=1) for k in range(3)]
covered = [inside[third == k].mean() for k in range(3)]

print(f"residual sd by income third: {spread[0]:.1f}, {spread[1]:.1f}, {spread[2]:.1f}")
print(f"inside the 95% band:          {covered[0]:.3f}, {covered[1]:.3f}, {covered[2]:.3f}")
```

[Run this code online](#)

45.1.3 Questions that need a distribution

Some questions cannot even be stated in terms of a conditional mean. *What share of a poor household's income goes on food?* asks for a low quantile, because policy is written about the vulnerable. *What is the tenth percentile of birth weight at this gestational age?* asks for a paediatric reference chart, a grid of conditional quantiles. *How much capital must be held against this portfolio?* asks for a value at risk, a conditional quantile and not a multiple of a conditional standard deviation unless the shape is known. *Has inequality in wages changed, or*

only the average wage? compares quantile curves at different levels, which is where quantile regression entered economics.

45.1.4 Three routes

The three sections that follow take three strategies, in increasing order of what they assume: estimate each quantile separately by an asymmetrically weighted absolute deviation; estimate each expectile separately by an asymmetrically weighted *squared* deviation, buying a differentiable loss at the price of a fitted quantity that is harder to describe; or model the whole conditional law by giving every parameter of a parametric family its own structured additive predictor. A fourth route was taken long ago — transform the response until a homoscedastic model fits, as in Definition 22.2.1 — but one power must serve the whole design, and Section 22.5 recorded what that costs when the answer is wanted on the original scale. Because the routes answer different questions, they are compared in Section 45.6 by which produces the better predictive distribution.

45.1.5 Exercises

45.1.5.1 A. Check your understanding

Exercise. A1

In a homoscedastic linear model with normal errors, by how much does the 0.9 conditional quantile exceed the 0.5 conditional quantile? Show that the answer does not depend on $\mathbf{x}$, and say which part of Proposition 45.1.2 this illustrates.

Solution. The difference is $\sigma\{\Phi^{-1}(0.9) - \Phi^{-1}(0.5)\} = \sigma\Phi^{-1}(0.9)$, with $\Phi^{-1}(0.9) \approx 1.2816$. By Proposition 45.1.2(a) the two quantile curves are the regression curve shifted by constants, so their difference is a constant free of $\mathbf{x}$.

45.1.5.2 B. Practice

Exercise. B1

Let $\log Y \mid x \sim N(\alpha + \beta x, \sigma^2)$ with σ constant. Show that the conditional quantiles of Y are *not* parallel, that the conditional coefficient of variation is constant, and that the conditional skewness is constant. Which parts of Proposition 45.1.2 apply to Y , and which to $\log Y$?

Solution. By Lemma 45.1.1 with $h = \exp$, $Q_Y(\tau \mid x) = \exp\{\alpha + \beta x + \sigma\Phi^{-1}(\tau)\}$, so the ratio of two quantile curves is constant while their difference is not: the curves fan out. $E(Y \mid x) = \exp(\alpha + \beta x + \sigma^2/2)$ and $\text{Var}(Y \mid x) = E(Y \mid x)^2(e^{\sigma^2} - 1)$, so the coefficient of variation is $(e^{\sigma^2} - 1)^{1/2}$, free of x ; the skewness of a lognormal depends only on σ . Thus $\log Y$ obeys part (a) and Y obeys part (b) with $\exp(\mathbf{x}^\top \boldsymbol{\alpha})$ proportional to $\exp(\beta x)$: a covariate-dependent scale but a frozen shape.

Exercise. B2

In [Example \(Engel's budgets, one more time\)](#) the homoscedastic 95% prediction interval has half-width close to 2×114.1 francs for every household. Using the within-third standard deviations, say roughly what nominal level that half-width actually attains in the poorest third and in the richest third, if the conditional law were normal with the within-third standard deviation. Comment on the direction of both errors.

45.1.5.3 C. Going deeper

Exercise. C1

[Proposition 45.1.2\(c\)](#) says that a location-scale model freezes the shape. The converse fails, and badly: freezing the conditional mean and variance leaves the shape free. Let Y given x take the value e^x with probability $1/(1 + e^{2x})$ and the value $-e^{-x}$ with probability $e^{2x}/(1 + e^{2x})$.

- (a) Show that $E(Y | x) = 0$ and $\text{Var}(Y | x) = 1$ for every x . A regression of Y on x with a constant variance is exactly right.
- (b) Show that $E(Y^3 | x) = 2 \sinh x$, so the conditional skewness runs over the whole real line as x does.
- (c) Show that

$$Q_Y(\tau | x) = \begin{cases} e^x, & \tau \leq 1/(1 + e^{2x}), \\ -e^{-x}, & \text{otherwise,} \end{cases}$$

and that for each fixed τ the curve $x \mapsto Q_Y(\tau | x)$ drops from e^{x_τ} to $-e^{-x_\tau}$ at $x_\tau = \frac{1}{2} \log\{\tau/(1 - \tau)\}$. At $\tau = 0.1$ the jump is from $\frac{1}{3}$ to -3 .

- (d) Show that the interval $[-2, 2]$, two conditional standard deviations either side of the conditional mean, has conditional coverage 1 when $|x| \leq \log 2$ and $e^{2|x|}/(1 + e^{2|x|})$ otherwise, so that its coverage ranges over $[\frac{4}{5}, 1]$ as x varies.
- (e) Say which part of [Proposition 45.1.2](#) this complements and why it contradicts none of them.

Solution.

- (a) Write $p = 1/(1 + e^{2x})$, so $1 - p = e^{2x}/(1 + e^{2x})$. Then $E(Y | x) = e^x p - e^{-x}(1 - p) = \{e^x - e^{-x}e^{2x}\}/(1 + e^{2x}) = 0$, and $E(Y^2 | x) = e^{2x}p + e^{-2x}(1 - p) = \{e^{2x} + 1\}/(1 + e^{2x}) = 1$.
- (b) $E(Y^3 | x) = e^{3x}p - e^{-3x}(1 - p) = \{e^{3x} - e^{-x}\}/(1 + e^{2x}) = e^{-x}(e^{4x} - 1)/(1 + e^{2x}) = e^{-x}(e^{2x} - 1) = e^x - e^{-x} = 2 \sinh x$, and since the variance is one this third moment is the skewness. It is 0 at $x = 0$ and unbounded in both directions.
- (c) The two atoms satisfy $-e^{-x} < 0 < e^x$ and the distribution function jumps to $1 - p$ at $-e^{-x}$ and to 1 at e^x . By the definition of Q_Y as an infimum, $Q_Y(\tau | x) = -e^{-x}$ when $\tau \leq 1 - p$ and e^x otherwise, which is the display with $p = 1/(1 + e^{2x})$. Solving $\tau = 1 - p = e^{2x}/(1 + e^{2x})$ gives $e^{2x} = \tau/(1 - \tau)$, that is $x = x_\tau$; for $x < x_\tau$ the quantile is the positive atom and for $x \geq x_\tau$ the negative one. At $\tau = 0.1$, $x_\tau = \frac{1}{2} \log \frac{1}{9} = -\log 3$, and the curve falls from $e^{-\log 3} = \frac{1}{3}$ to $-e^{\log 3} = -3$.
- (d) Both atoms lie in $[-2, 2]$ exactly when $e^x \leq 2$ and $e^{-x} \leq 2$, that is when $|x| \leq \log 2$, and then the coverage is 1. For $x > \log 2$ only the atom e^x is outside, so the coverage is $1 - p = e^{2x}/(1 + e^{2x})$; for $x < -\log 2$ only $-e^{-x}$ is outside and the coverage is $p = e^{2|x|}/(1 + e^{2|x|})$. The smallest value is at $|x| = \log 2$, namely $4/(1 + 4) = \frac{4}{5}$, and the coverage returns to 1 as $|x| \rightarrow \infty$.

- (e) It complements part (c), which shows that a location-scale family has a shape free of $\mathbf{x}$; the present family shows that the implication does not reverse, so that a correct mean model and a correct variance model together still say nothing about skewness, about quantiles, or about the coverage of an interval built from them. Nothing is contradicted, because this conditional law is not of the form $\mathbf{x}^\top \boldsymbol{\beta} + \exp(\mathbf{x}^\top \boldsymbol{\alpha})\varepsilon$ for any ε free of $\mathbf{x}$ — that is the content of [Exercise \(C2\)](#). The band of [Example \(Engel's budgets, one more time\)](#) behaved in the same way, covering 1.000 of the poorest third and 0.911 of the richest while reporting one number for all.

Exercise. C2

The fourth route mentioned at the end of the section is to transform the response until a location-scale model fits. Here it provably cannot work.

- (a) Suppose T is strictly increasing and left-continuous and $T(Y) = a(x) + b(x)\varepsilon$ with $b(x) > 0$ and the law of ε free of x . Use [Lemma 45.1.1](#) to show that $Q_Y(\tau | x) = T^{-1}\{a(x) + b(x)q_\varepsilon(\tau)\}$, and deduce that for any three levels $\tau_1 < \tau_2 < \tau_3$ the ratio

$$\frac{T\{Q_Y(\tau_3 | x)\} - T\{Q_Y(\tau_2 | x)\}}{T\{Q_Y(\tau_2 | x)\} - T\{Q_Y(\tau_1 | x)\}}$$

is the same for every x . This is the constraint a transformation imposes, and it is checkable.

- (b) Show that if Z takes two values $\alpha < \gamma$ with $\Pr(Z = \gamma) = q \in (0, 1)$, then $\{Z - E(Z)\}/\sqrt{\text{Var}(Z)}$ takes the values $-\sqrt{q/(1-q)}$ and $\sqrt{(1-q)/q}$ with probabilities $1-q$ and q : the standardized law is determined by q alone.
- (c) Deduce that for the family of [Exercise \(C1\)](#) no strictly increasing T makes $T(Y)$ a location-scale family in x , and hence that no Box–Cox transformation (Definition 22.2.1) can homogenize it.
- (d) What is left, and where does the chapter go next?

Solution.

- (a) Applying [Lemma 45.1.1](#) conditionally on x to the map $e \mapsto a(x) + b(x)e$ gives $Q_{T(Y)}(\tau | x) = a(x) + b(x)q_\varepsilon(\tau)$, and applying it to T itself gives $Q_{T(Y)}(\tau | x) = T\{Q_Y(\tau | x)\}$. Hence $T\{Q_Y(\tau | x)\} = a(x) + b(x)q_\varepsilon(\tau)$; the differences across levels cancel $a(x)$ and the ratio cancels $b(x)$, leaving $\{q_\varepsilon(\tau_3) - q_\varepsilon(\tau_2)\}/\{q_\varepsilon(\tau_2) - q_\varepsilon(\tau_1)\}$, free of x .
- (b) $E Z = \alpha + q(\gamma - \alpha)$ and $\text{Var } Z = q(1-q)(\gamma - \alpha)^2$, so $\{\gamma - E Z\}/\sqrt{\text{Var } Z} = (1-q)(\gamma - \alpha)/\{(\gamma - \alpha)\sqrt{q(1-q)}\} = \sqrt{(1-q)/q}$ and likewise for α . Neither the location nor the scale of Z survives, only q .
- (c) A strictly increasing T maps the two atoms of Y to two atoms in the same order and leaves their probabilities alone, so $T(Y)$ given x is two-point with $\Pr\{T(Y) = T(e^x)\} = 1/(1+e^{2x}) =: q(x)$. If $T(Y) = a(x) + b(x)\varepsilon$ with $b(x) > 0$, then $\varepsilon = \{T(Y) - a(x)\}/b(x)$ has a law free of x , so in particular its standardized version does; by (b) that forces $q(x)$ to be constant. But $q(x) = 1/(1+e^{2x})$ is strictly decreasing. Hence no such T exists, and the Box–Cox family, being a family of strictly increasing maps on a half-line, contains none either.
- (d) What is left is to stop insisting that one curve with one noise distribution describes the conditional law. Either estimate each conditional quantile on its own terms, as

Section 45.2 does with the check loss, or give every parameter of a distribution its own predictor, as Definition 45.5.1 does. The family here is an extreme case built to make the point; the Engel data of Example (Engel's budgets, one more time) make a milder version of it, with a spread that triples and a skewness that changes sign across the design.